

L3-2 | Frozen Judgment: A Coordinator's Record

In engineering projects, not all failures originate from the absence of judgment. Some issues are identified early, discussed, and even addressed through corrective actions. They enter formal processes and are, at some point, considered “resolved.”

The real risk often emerges after that moment.

Once a problem is formally taken over, assigned a handling path, and confirmed through interim results, it simultaneously loses the space to be redefined. Judgment no longer revolves around the problem itself, but around maintaining stability through existing conclusions and established procedures. As long as no explicit failure is observed, the original judgment continues to be applied.

In such a state, the problem is not ignored; it is frozen.

It no longer triggers cross-disciplinary discussion, nor does it return to the level of system assumptions for re-examination. Any further questioning is treated as repetitive, beyond authority, or unjustifiable in terms of time and cost. Judgment is completed in form, but ceases to evolve in substance.

This chapter focuses precisely on such situations: problems that once entered the judgment system and were confirmed as “closed,” yet whose systemic implications did not disappear as a result. When external conditions change, the system can only expose the boundary of that judgment through incidents.

What follows is not a technical failure review, but a record of how a judgment, once completed, loses its ability to be judged again.

Project Background (Factual Summary)

- Time: Summer 2019

Geographic context: A coastal city in Guangdong Province

Project subject: Renovation of an existing comprehensive hospital building

- Building characteristics:
 - Single building
 - Nine stories above ground
 - Existing structure, not new construction
- Renovation scope:
 - Limited to the 8th and 9th floors
 - Other floors retained original functions
- Original function:
 - Administrative and auxiliary spaces on the 8th and 9th floors
- Post-renovation functions:
 - 8th floor: Imaging and comprehensive laboratories
 - General testing
 - Imaging-related spaces
 - PCR-related areas
 - 9th floor: Oncology-oriented biological laboratory
 - Experimental subjects: mice
- Building conditions:
 - Typical floor height approximately 3.2 m
 - Primary structural system unchanged
- Roof conditions:
 - Flat roof
 - Micro-slope drainage, approximately 3‰
 - Parapet height approximately 1.6 m
 - No centralized mechanical penthouse
- Roof-mounted equipment:
 - Cooling towers
 - Fresh air handling units
 - Exhaust air units
 - Pumps
 - Newly added VRF outdoor units (3 sets)
- System renovation highlights:
 - Additional fresh air and exhaust systems to support new laboratory functions
 - Multiple new duct penetrations through roof and top floor slab
- Penetration characteristics:
 - Post-installed openings
 - Serving new ductwork
 - No new equipment foundations involved
 - Penetrating waterproofing layer, leveling layer, and structural slab
- Project status prior to incident:
 - Systems in commissioning phase
 - Cleanroom and pressure differential systems accepted
 - Project in delivery status

Incident Record

Initial Anomaly

After completion of major construction activities, the project entered system commissioning and functional verification. Cleanroom and pressure control systems were accepted in accordance with established procedures, and relevant documentation was archived. Portions of the 8th floor were operational, while the 9th floor had not yet been formally put into use. Project personnel remained on site for routine inspection and close-out coordination.

During a routine inspection, site personnel observed abnormal traces on the floor of the clean fresh air equipment room on the 8th floor. The location was directly beneath a major duct penetration through the top floor slab. Localized white crystallization and minor residual moisture were observed. At the time, the area was not under cleaning or washdown operations, and no evidence of manual water discharge or equipment drainage was found nearby.

Upon inspection, the area had previously undergone waterproofing repair and passed a stage inspection. Equipment was operating normally at the time of discovery, and no immediately identifiable source of leakage was observed. The condition was documented with photographs and logged in the daily inspection record.

- The information was reported to project coordination parties, and further investigation and observation regarding the leakage source were initiated.

Initial Response and Stage Closure

Following documentation of the initial anomaly, relevant parties organized investigation and corrective actions. The focus was on the roof structure, waterproofing layer, and newly added duct penetrations. Based on observable conditions at the time, the anomaly was preliminarily attributed to localized deficiencies in waterproofing execution.

Targeted waterproofing repair measures were implemented, including reinforcement and sealing of the affected areas. Post-repair inspection did not reveal immediate leakage.

The project then entered an observation period. Over approximately three weeks, no new anomalies were recorded. During this period, climatic conditions were relatively mild, with average temperatures between 22–26°C, no significant precipitation, and no sustained strong solar exposure.

At the conclusion of the observation period, corrective actions were formally confirmed and signed off according to established procedures. A stage conclusion was formed indicating the anomaly had been addressed, and the project proceeded with delivery activities.

Subsequent Anomaly

After completion of stage corrective actions and entry into delivery status, the affected areas were no longer treated as priority observation points.

During a later routine inspection, another leakage anomaly was observed. This occurrence was located above an animal housing room on the 9th floor, where water penetrated through a color steel ceiling panel and dripped into the clean space below. This location differed from the previously recorded anomaly on the 8th floor and was not within the vertical projection of the original penetration.

Inspection confirmed that no new equipment foundations were present above the ceiling, no recent construction activities had occurred nearby, and indoor equipment was operating normally. The leakage was documented and reported.

At the time of this second anomaly, the 8th floor laboratory areas had completed cleanroom and pressure acceptance and were in use. The 9th floor had not yet been formally commissioned. The project as a whole was already in delivery status, and related corrective items were institutionally closed.

Due to the shift in leakage location and its separation from the original repair area, the anomaly could not be directly classified as a continuation of the initial incident, prompting further attention.

L3-2 Stage Judgment and Positional Statement

It must first be stated that the events recorded in this chapter did not conclude at this point. The second anomaly, appearing as an incident, merely forced the boundary of the existing judgment to become visible again; it did not automatically trigger system-level redefinition or path restructuring. Related issues continued to exist in subsequent phases in different forms, and their further development lies beyond the scope of this chapter. The decision to stop here does not indicate that the problem was resolved, but that, as a judgment object, it has fulfilled the responsibility required to support this chapter's objective. Subsequent developments and longer-term impacts will be addressed in later chapters as separate analytical dimensions.

Throughout the process recorded here, the issue was not in a state of loss of control. It was identified, discussed, handled, and confirmed through established procedures. From an institutional and procedural perspective, the judgment chain was complete. For this reason, the second anomaly did not point to a specific technical error. What it exposed was not a missed construction detail, but a more difficult question for the system to recognize: whether the system retains the ability to redefine a problem once a judgment is considered complete.

Unlike the grey zones discussed in L3-1, the situation presented here did not result from responsibility avoidance or failure to take over the problem. On the contrary, the issue had been clearly incorporated into the judgment system and appeared to be validated during a period of stability. Judgment did not fail because it was ignored, but because it was prematurely sealed. The incident functions as a counter-evidence not because of its severity, but because it occurred at a point when the system had already defined the project as “operable,” “deliverable,” and “ready for judgment closure.”

In this process, the role of the coordination interface was neither to propose solutions nor to lead technical decisions. Its value lay solely in maintaining continuity of information: which conditions had been used to support the judgment, which changes had not been incorporated into its premises, and why, when the anomaly reappeared, the system no longer possessed an interface to return to the level of assumptions for correction. When the incident emerged as counter-evidence, the system did not immediately re-enter a state of reassessment; instead, it first sought to determine whether the existing conclusion could still be maintained. This response itself constitutes the final manifestation of frozen judgment.

Subsequent chapters will shift to another type of situation: cases in which judgment freezing does not surface through a single incident, but persists as gradual deviation, responsibility diffusion, or deferred risk, and how such conditions are observed from a coordination perspective and handled by systems over longer time scales.

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